Abinaya G
Department of Information Technology
Saveetha Engineering College
Chennai, India

November 10, 2026

The Editors-in-Chief
Expert Systems with Applications

Dear Editors,

I submit the manuscript *"EIG-MPHBS: a rotation-invariant information exchange for hybrid Honey Badger–Marine Predators optimisation"* for consideration as a research article in *Expert Systems with Applications*.

Hybrid metaheuristics are widely used as the optimisation engine of intelligent systems in engineering. Many of them let two populations cooperate by recombining their members coordinate by coordinate. The manuscript shows that this exchange depends on the coordinate system and fails on problems whose variables are coupled. It does so for MPHBS, a recent hybrid of the Honey Badger Algorithm and the Marine Predators Algorithm (Flores and Olivares, *Knowledge-Based Systems* 352, 2026). It then fixes the weakness by carrying out the exchange in the eigenvector frame of the pooled populations.

The manuscript does not propose a new metaphor-based algorithm. The method is stated in standard optimisation terminology (covariance estimation, eigenvector frame, coordinate-wise recombination). It is positioned against the eigenvector-based crossover and multitasking literature it builds on, and its properties are proved formally. The main contributions are:

- a proof that the eigen-frame exchange is invariant to rotations of the search space and costs no extra function evaluations, with a measured run-time overhead of 3–6%;
- a mechanism analysis showing that the eigen frame restores the acceptance of exchanged candidates on coupled problems, where the axis-frame exchange stalls;
- an evaluation in 18 840 seed-paired runs on CEC2022, the 24 BBOB functions and eight FIR filter-design problems against ten algorithms: MPHBS, its parent algorithms, PSO, DE, GWO, WOA, L-SHADE and CMA-ES, plus the official CEC2022 results of EA4eig;
- a design fixed before the experiments, with confirmatory tests on held-out instances, and public code, raw runs and analysis scripts.

The results are reported in full, including their limits. EIG-MPHBS is never significantly worse than MPHBS on any of 56 instances and clearly outperforms the classical metaheuristics, but L-SHADE, CMA-ES and EA4eig remain stronger. The FIR filter-design problems provide the engineering application.

The manuscript cites one related manuscript by the author, which is under review elsewhere. It is cited anonymously as "Anonymous (2026)" and identified on the title page; it studies the learned mediator of MPHBS and does not overlap with the method or experiments of this submission. I am glad to provide a copy if useful.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. The author declares no competing interests, and the research received

no specific funding.

Thank you for considering this submission.

Yours sincerely,

Abinaya G
Department of Information Technology
Saveetha Engineering College,
Chennai, India
gabinaya@saveetha.ac.in